

Exponentials and logarithms - Set 1

TMUA practice questions by topic. Specification MM5.1-MM5.3. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

1 Find the value of $\log_2 40 - \log_2 5 + \log_2 \frac{1}{4}$.

- A 1
- B 2
- C -1
- D 5
- E 3
- F 0

2 Solve $3^{2x} = 4$. Give x in terms of logarithms to base 3.

- A $x = \log_3 4 - 2$
- B $x = \log_3 2$
- C $x = \frac{\log_3 4}{4}$
- D $x = \log_2 3$
- E $x = \frac{1}{2} \log_3 2$
- F $x = 2 \log_3 2$

3 Solve $2^{2x} - 3 \cdot 2^{x+2} + 32 = 0$. Find the sum of the solutions.

- A 6
- B 12
- C 4
- D 3
- E 8
- F 5

- 4 Given that $\log_a x = 3$ and $\log_a y = 5$, find $\log_a \left(\frac{x^2 \sqrt{y}}{a} \right)$.
- A 8
 B $\frac{15}{2}$
 C $\frac{13}{2}$
 D $\frac{17}{2}$
 E 10
 F $\frac{11}{2}$
- 5 Solve $\log_3 (x + 6) + \log_3 (x - 2) = 2$.
- A $x = 5$
 B $x = -7$ or $x = 3$
 C $x = 3$
 D $x = 3$ or $x = 7$
 E $x = -3$
 F $x = 1$
- 6 Which of the following statements is/are true for all positive real x and y ?
- I $\log_{10} (x + y) = \log_{10} x + \log_{10} y$
 II $\log_{10} (xy)^2 = 2\log_{10} x + 2\log_{10} y$
 III $\log_{10} x - \log_{10} y = \log_{10} x \div \log_{10} y$
- A none of them
 B I only
 C II only
 D III only
 E I and II only
 F I and III only
 G II and III only
 H I, II and III

- 7 Find the value of $\frac{\log_5 27}{\log_5 3}$.
- A 9
B $\frac{27}{3}$
C $\log_5 9$
D 3
E $\log_5 24$
F 24
- 8 How many real solutions does the equation $4^x + 2^{x+1} - 8 = 0$ have?
- A 1
B 2
C 0
D 3
E infinitely many
- 9 Solve $\log_2 x + \log_4 x = 6$. (You may use $\log_4 x = \frac{1}{2}\log_2 x$.)
- A $x = 8$
B $x = 2^6$
C $x = 16$
D $x = 64$
E $x = 4$
F $x = 32$
- 10 The curve $y = a^x$ passes through (2, 12) and (4, 108). Find a .
- A 4
B 3
C $\sqrt{12}$
D 9
E 2
F 6

11 Given that $\log_{10} 2 = p$ and $\log_{10} 3 = q$, express $\log_{10} 0.75$ in terms of p and q .

A $3q - 4p$

B $\frac{q}{2p}$

C $2p - q$

D $q + 2p$

E $q - p$

F $q - 2p$

12 Solve $\log_2 (x^2 - 4) - \log_2 (x - 2) = 3$.

A $x = 2$

B $x = 6$

C $x = -10$ or $x = 6$

D $x = 8$

E $x = 10$

F $x = 4$